Experiment: -8

Write MATLAB programs for generating the following continuous time signals and plot them using subplot command. Also, make sure to use: x-Label, y-label, Axis, Grid, hold on, Title appropriately for each and every plot. Use HELP for understanding any unknown function.

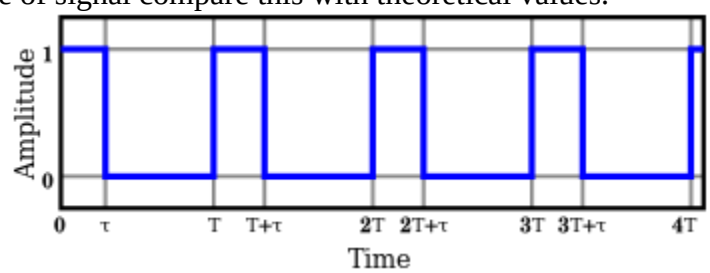
1. Fourier Series

(A) I. Consider the pulse signal shown below. Write the expression of Trigonometric Fourier series coefficients Fourier series in your notebook.

II. Plot the Fourier coefficient of the pulse waveform with T=1,τ =0.25 using MATLAB.

III. Plot the amplitude and phase spectrum of the Fourier coefficient.

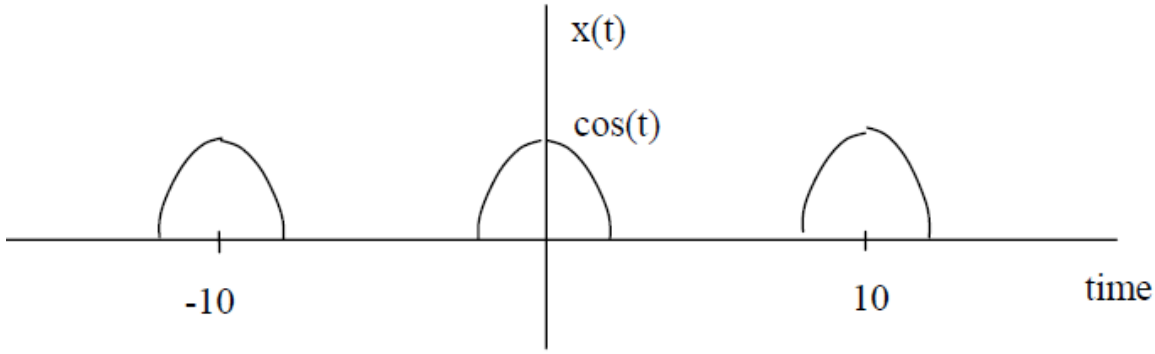
IV. Find out the dc amplitude, fundamental frequency amplitude and other harmonic (till 4th) component amplitude of signal compare this with theoretical values.



Reference Program For Problem (A): -

```
clc
clear all;
Xl=1;
T=1;
Tl=0.25;
A0_2=Xl*(0.25)/T;
k=1:18;
Ak=Xl*sin(2*pi*k*0.25/T)./(k*pi);
Bk=Xl*(1-cos(2*pi*k*0.25/T))./(k*pi);
Thk=atan2(Bk,Ak);
Mk=sqrt(Ak.*Ak+Bk.*Bk);
fk=k/T;
subplot(211)
stem([0 fk],[A0_2 Mk])
xlabel('Frequency - Hz')
ylabel('Amplitude/X1 - V')
title('AMPLITUDE SPECTRUM')
subplot(212)
stem([0 fk],[0 Thk])
xlabel('Frequency - Hz')
ylabel('Phase - rads.')
title('PHASE SPECTRUM')
```

(B) For the following signal



I. Find the exponential Fourier series coefficients using MATLAB program for n = 15.
II Plot the spectra versus frequency.
III. Plot the amplitude and phase spectrum of the Fourier coefficient.
IV. Find out the dc amplitude, fundamental frequency amplitude and other harmonic (till 4) component amplitude of signal compare this with theoretical values.

(C) I. Consider a triangular wave of period 1sec, amplitude of 2Vpp and no DC value. Write the expression of Trigonometric Fourier series coefficients Fourier series in your notebook.
II. Plot the Fourier coefficient of the waveform using MATLAB.
III. Plot the amplitude and phase spectrum of the Fourier coefficient.
IV. Find out the dc amplitude, fundamental frequency amplitude and other harmonic component amplitude of signal compare this with theoretical values.

2. Gibbs Phenomenon

(A) Fourier series are used to approximate complex functions. By summing sine waves it is possible to replicate the complex function almost exactly, however, there is a discontinuity in this periodic function, meaning the Fourier series will also have a discontinuity.
Square wave expression:-

$$F(t) = \begin{cases} -1 \; for -1 \leq t < 0 \\ 1 \; for \; 0 \leq t \leq 1 \end{cases}$$

Fourier Sum of square wave can be represented as given below.

$$F(t) = \frac{4}{\pi} \sum_{j=odd}^{\infty} \frac{1}{j} sin(j\pi t)$$

Write a MATLAB code to plot the square wave function along with the Fourier series in order to compare them for different value of “j”.

(B) Consider the signal of Question 1 Part (B and C) and regenerate the original signal from the Fourier components and observe Gibbs Phenomena for n = 10, 20 and 50.